SPL3 Project Proposal Form, 2026

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Project Title: **General Purpose Transfer Learning Framework for Cross-Domain Classification**

Motivation: Traditional machine learning models generally require large amounts of **labeled** training data with distributions similar to the target data. In many real-world applications, collecting sufficient labeled data is expensive, time-consuming, and often impractical. The challenge becomes even more severe in **cross-domain environments** due to annotation scarcity, domain mismatch, and the lack of sufficient labeled target-domain data. As a result, conventional machine learning approaches often fail to generalize effectively across domains.

Transfer learning addresses this issue by **transferring knowledge** between related domains, reducing the dependence on **large task-specific datasets**. However, transfer learning itself may suffer from negative transfer, where knowledge from a source domain negatively impacts target-domain performance because of low domain similarity or ineffective adaptation.

Therefore, this project is motivated by the need for a reusable and extensible transfer learning framework capable of analyzing domain relationships, reducing negative transfer, and **improving performance in low-resource and cross-domain** scenarios.

Project Description:

Overview: This project proposes a modular transfer learning framework designed for cross-domain classification tasks across text, and image datasets. The framework will provide unified data preprocessing, feature transformation, model training, evaluation, and visualization components within a single platform.

The system will support multiple transfer learning and domain adaptation techniques, including TCA[1], JDA[2], BDA[3], CORAL[6],DALI[5], GFK, JCSL,SDA, OCMTL[4] and my proposed enhanced method OCMTL+, enabling comparative analysis under different domain shifts and data availability conditions. It will also include experiment tracking, configurable hyperparameters, and interactive visualization for analyzing feature distributions and adaptation effectiveness.

The application includes text analysis (sentiment analysis), and image recognition. The expected outcome is a **scalable** and reusable **framework** that simplifies cross-domain experimentation, reduces training effort, and improves learning performance in data-scarce environments.

Project Scope and Features

1. Unified Data Processing Pipeline

- Centralized workflow for data loading, preprocessing, normalization, feature preparation, and train/test splitting.
- Support for configurable preprocessing and normalization strategies.

2. Multi-Domain Dataset Support

- Support for text, image, and software datasets.
- Compatibility with multiple file formats (.csv, .mat, .arff, etc.).
- Custom source-target domain tagging system for transfer learning experiments.

3. Transfer Learning and Baseline Modeling

- Model training for comparative analysis.
- Cross-domain transfer learning evaluation framework.
- Modular architecture for integrating new transfer learning methods

4. Performance Evaluation Framework

- Evaluation using Accuracy, AUC, Precision, Recall, and F1-Score.
- Comparative performance analysis across multiple methods.
- Detection and analysis of negative transfer scenarios.

5. Interactive User Interface and Dashboard

- Dataset upload and experiment configuration.
- Transfer method selection and hyperparameter customization. (regularization, latent features, kernel type, normalization and standardization type)
- Real-time visualization and comparison of results.
- Data property inspection before (sample, feature, datatype) and after transfer (MMD).
- Exportable reports in PDF and CSV formats.

6. Visualization and Interpretability

- Feature space visualization using PCA and t-SNE.
- Domain alignment and scatter plot visualization.

7. Cross-Domain Transfer Analyzer

- Domain similarity analysis using MMD and feature distribution overlap.
- Identification of beneficial vs harmful transfer scenarios.
- Recommendation system for selecting optimal transfer methods.

Languages or Tools to be used:

Category	Details
Programming Languages	Python, , PyQt
Supporting Libraries	Scikit-learn (models), Pandas & Numpy (data processing), Matplotlib / Seaborn (result visualization);
Transfer Learning Methods	TCA, JDA, BDA, CORAL, OCMTL, DALI, JFK, JCSA, OCMTL+ (proposed method by me)
Dataset	COIL20, MNIST, PIE, Office-caltech (Surf), Office-caltech (Decaf), Office-31 (Decaf), Reuters
Tools	Git, GitHub (Version Control); VS code, Jupyter Notebook, Matlab (Code Editor)

Reference:

1. Pan, Sinno Jialin, et al. "Domain adaptation via transfer component analysis." *IEEE transactions on neural networks* 22.2 (2010): 199-210.
2. Long, Mingsheng, et al. "Transfer feature learning with joint distribution adaptation." *Proceedings of the IEEE international conference on computer vision*. 2013.
3. Wang, Jindong, et al. "Balanced distribution adaptation for transfer learning." *2017 IEEE international conference on data mining (ICDM)*. IEEE, 2017.
4. Nahid, Shartaz Sajid, et al. "OCMTL: Transfer learning by orthogonal core-extraction of a matrix and its application in cross-project defect prediction." *Alexandria Engineering Journal* 133 (2025): 498-516.
5. Devika, A. K., et al. "Visual domain adaptation through locality information." *Engineering Applications of Artificial Intelligence* 123 (2023): 106172.
6. Sun, Baochen, Jiashi Feng, and Kate Saenko. "Correlation alignment for unsupervised domain adaptation." *Domain adaptation in computer vision applications*. Cham: Springer International Publishing, 2017. 153-171.

Supervisor's Name: Dr. Mohammad Shoyaib

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation 1:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Proposal Presentation Feedback (if any):

Midterm 1 Presentation Feedback (if any):

Midterm 2 Presentation Feedback (if any):